Section: 03

Group No: 07

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Resistivity

Equipment

Included:		
1	Resistance Apparatus	EM-8812
1	Voltage Sensor	UI-5100
2	Patch cords	SE-9750
NOT INCLUDED, BUT REQUIRED:		
1	550 Universal Interface	UI-5000
1	Micrometer	SE-7337

Introduction

The resistivity of different metals is determined by finding the resistance of wires of a known diameter as a function of their length. It is also shown that the resistance of a wire of fixed length is inversely proportional to its cross-sectional area.

Written by Chuck Hunt

Theory

If a current (I) is flowing through a wire, the voltage drop (V) across a certain length of wire with resistance R is given by Ohm's Law:

$$V=IR$$

or solving for R gives:

$$R = V/I \qquad (1)$$

In this experiment, you will measure V and I to determine R for various lengths of wire. You will then make a graph of R versus length L. The resistance of a wire depends on the length of the wire, the cross-sectional area, A, and the resistivity, ρ, of the material:

$$R = \rho L/A \qquad (2)$$

A plot of R vs. L will result in a straight line that has a slope equal to ρ/A.

Thus the resistivity is given by

$$\rho = (\text{slope})A = (\text{slope})\pi(d/2)^2 \qquad (3)$$

where d is the diameter of the wire.

Setup

1. Make the connections as shown in Figure 1.

2. Select four brass wires of different diameter and measure* their diameters. Also select Nichrome wire and measure the diameter. Brass is yellow in color. The Nichrome is a uniform dark g
Change the values in the Brass Wire table under the Data tab to match your values for Bra
Change the values in the Different Metals table under the Resistivity tab to match your valor the two different metals.

   *If you do not have a micrometer (or good digital calipers), use these values of diameter:

   0.127 cm
   0.101 cm (use this for the non-brass wires as well)
   0.082 cm
   0.051 cm

3. On the Resistivity apparatus, move the Reference Probe and the Slider Probe to the Park pion. The probes should be as far left and right respectively as possible so the probe lifts up to a installation of the sample wire. They will click into position.

4. Turn the two black handles counterclockwise to open the clamps to allow the sample wire tide into position.

5. Install the Nichrome wire in the apparatus. Slide from left or right using the white line-up hash marks. Figure 2 shows the right hand side as the wire slides in. Note that on the right hande, the wire is on the far side of the silver clamp (with black handle), but on the left hand side vire will be on the near side of the clamp as shown in Figure 3. This prevents the wire from bov as you tighten the clamps.

6. Tighten the clamps by turning the black handles clockwise.

7. Position the reference probe at the 0 cm mark and the slider probe at the 5 cm mark.

Figure 1: Apparatus Connections

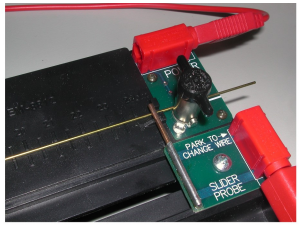

Figure 2: Right Hand Clamp

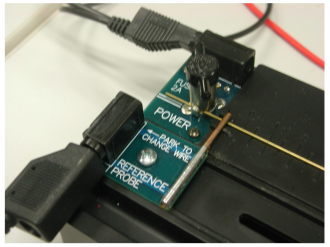

Figure 3: Left Hand Clamp

Procedure

1. Click open the Signal Generator (left side of screen). 550 Output should be set for a DC voltage of 2.0 V. Click the On button to turn the Sigr Generator on.

2. Click the Data Summary button at the left of the screen. Click the Gear icon on the top right of the two rows that say Voltage, Ch A. The Gai should be 10x. If not, click the white triangle and select 10x. Click OK.

3. Move the Slider Probe on the Resistance Apparatus so the contact is at 5.0 cm.

4. Read the rest of this page, then click open the Data tab. The program will ask if you wish to turn off the Signal Generator, click Leave On.

5. Click RECORD at the bottom left of the screen. Wait a few seconds until the numbers stop changing and then click STOP.

6. The resistance in the lower box is calculated from R = V/I where the V and I values are averages that show in the upper two boxes. In the fir row of Table I, enter the Resistance (first) and then the Wire Length (5.0 cm).

7. Repeat steps 5 & 6 for Slider Probe positions of 10.0 cm, 15.0 cm, 20.0 cm, & 24.0 cm. If you use the same positions for each run, you will on need to enter them for the Nichrome runs.

8. The Data summary should show five runs. Double click on the last run (probably Run #5) and re-label it Copper. Delete all the other runs us the white triangle by the Delete Last Run icon at the bottom of the page.

9. Repeat for the 1.0 mm (second largest) brass wire except do step 2 and change the Gain to 100x. When you do the 24 cm run, enter the Resistance in the Different Metal table as before, but also enter it in Table II in the 1 mm row.

10. Repeat for the other three brass wires, except only do the 24 cm position and enter the data in the Brass Wires table. Label the runs Brass 1 Brass 82, Brass 51.

Voltage

V (mV) Brass wire

1.40mV

Current

I (mA) Brass wire

16mA

Resistance

R (mΩ) Brass wire

88.51mΩ

Table I: Different Lengths

	Brass wire	Brass wire
	Wire Length (cm)	Resistance (mΩ)
1	5.0	4.70
2	10.0	8.88
3	15.0	12.98
4	20.0	16.86
5	24.0	20.26
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Table II: Brass Wires

	Resistance. (mΩ)	diameter (mm)
1	13.55	0.13
2	20.26	0.10
3	33.13	0.08
4	88.51	0.05
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Analysis

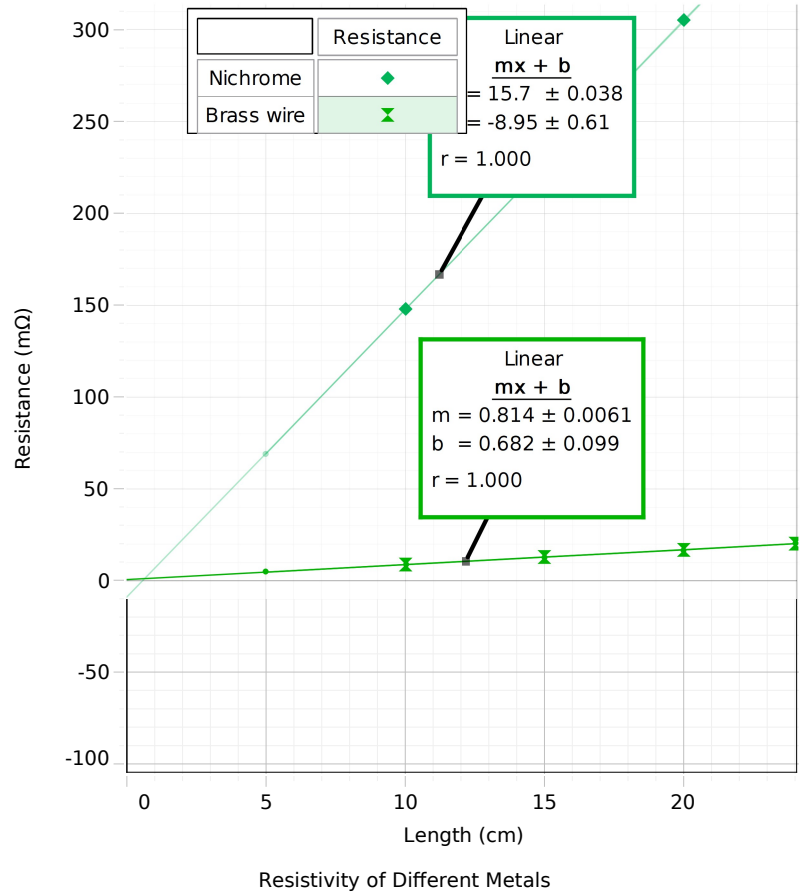
1. The graph shows the resistances you measured versus the lengt of wire you used. From the graph toolbar, click the Run Select black triangle and select the Copper run.

2. Click the Scale-to-Fit icon at the top left of the toolbar.

3. Click the black triangle by the Curve Fit icon on the toolbar and select Linear. Click anywhere on the graph to get rid of the black box. Right click in the Linear box and click on Show Uncertainty i it is not already showing.

4. Record the slope, m, of the R versus L graph in the Slope column of Table III under the Resistivity tab. Note that in most cases the uncertainty in the slope is less than 1%.

5. Repeat for each of the different metals.



Resistivity of Different Metals

Table III: Different Metals

Metal	Wire Diameter (cm)	Slope (mΩ/cm)	ρ (μΩ cm)	Resistivity (μΩ cm)
Brass	0.101	0.814	6.52	7.0 ± 0.5
Nichrome	0.101	15.7	126	105 ± 5

Conclusions:

1. How well does the data fit straight lines (see the Graph page)? Wh does this show?

We did not caliprated after assigning the nichrome wire. For this, the line for nichrome does not start from 0. The graph shows that resistence is proportional to length. Resistence of nichrome wire is much higher than the resistence of brass wire.

2. Click open the Calculator at the left of the page and verify that line 5 calculating ρ is in agreement with Equation 3 from Theory. The factor of 10 arises from converting from mm^2(mΩ)/cm to (μΩ)cm. Click the Calculator closed.

3. The value calculated for ρ is given in column 3. Except for Nichrome, the major source of uncertainty is the measurement of the diameter. If you measured this with a micrometer or digital calipers that read to 0.01 mm, the uncertainty in ρ is about 1%. Nichrome may have had a larger uncertainty in Slope due to spread in the data.

4. Column 5, labeled Resistivity, contains the manufacturer's values. The uncertainty arises because the metal wires are actually alloys and the actual resistivity depends on the exact composition.

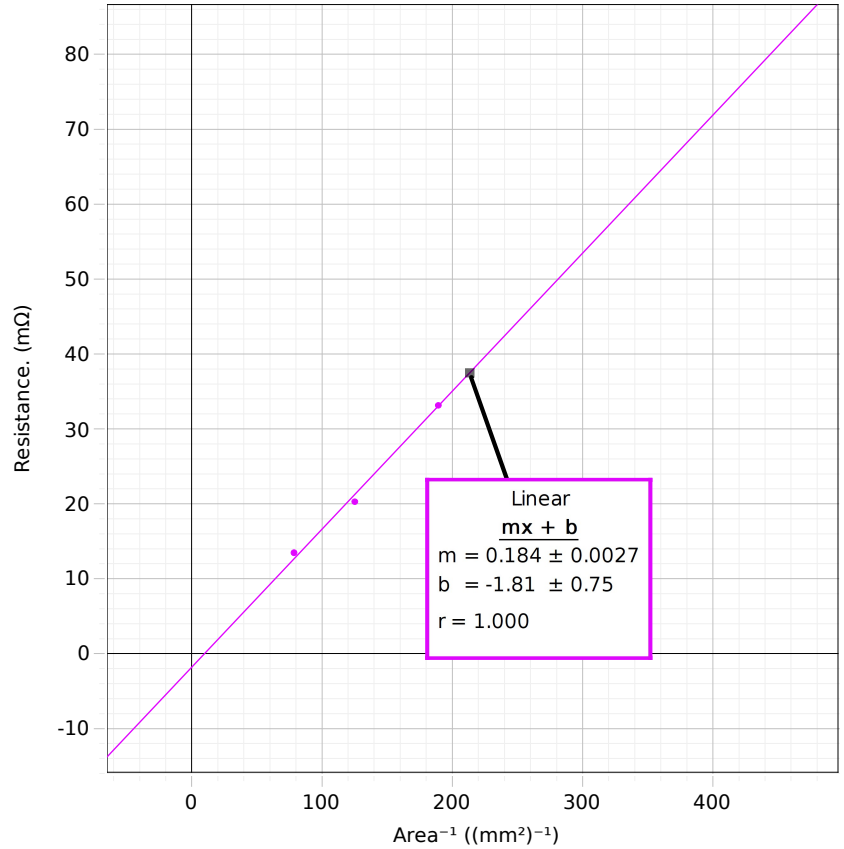
5. Discuss how well you data agrees with the given values. What does this show about Equation 3?

For nichrome wire my given data does not agrees with the given value and for brass wire it does agrees with the value. The data shows that the resistivity depends on materials.

Area Dependence

1. The graph shows the resistances you measured versus the cross-sectional area for the four brass wires. On the Area axis, click on the measurement and select the quick-calc[1]y
2. Click on the Curve Fit black triangle and turn on Linear.
3. How well does your data fit a straight line? What does this show about Equation 2 from Theory? What is the physical meaning of the slope?

My data fits the the straight line well. The graph shows that the cross section area is inversely proportional to the resistence which matches with the theory 2. The slope indicates that the cross section area is inversely proportional to the resistence.



Dependence of Resistance on Cross-sectional Area